# FUNCTIONAL IMPROVEMENT

## MOVING BEYOND DYSTROPHIN IN DMD



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# Speakers



## Aravindhan Veerapandiyan, MD

Associate Professor of Pediatrics
Director, Comprehensive Neuromuscular Program
Co-Director, MDA Pediatric Care Center
Arkansas Children’s Hospital



## Susan Apkon, MD

Fischahs Chair in Pediatric Rehabilitation
Children's Hospital Colorado
Professor and Executive Vice-Chair, Department of
Physical Medicine and Rehabilitation
University of Colorado School of Medicine

# Disclosures

## Aravindhan Veerapandiyan

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## Susan Apkon

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# Resetting expectations for functional improvement

Aravindhan Veerapandiyan, MD

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# Dystrophin quantity, quality, and distribution underlies functional improvement in DMD



## Beyond dystrophin: functional improvement in practice

- Preserving motor function and retaining independence are important to individuals with DMD and caregivers[1]
- Dystrophin deficiency manifests in a multisystemic manner – individuals can experience CNS, GI, bladder, cardiac, and respiratory manifestations[2]

## Dystrophin: quantity, quality, and distribution

- Increased dystrophin production
- Production of high-quality dystrophin
- Broad and effective distribution of dystrophin restorative treatments to affected muscles (skeletal, smooth, and cardiac) and the CNS

CNS, central nervous system; DMD, Duchenne muscular dystrophy; GI, gastrointestinal.

1. Gillman A, et al. *Front Neurol. 2025;*16:1509174; 2. Ohlendieck K, Swandulla D. *Pflügers Arch.* 2022;473:1813–1839.

# How is meaningful change in functional improvement defined?

## In clinical trials…

**MCID**
**Minimal clinically important difference**

The smallest difference that reflects a change that is meaningful for an individual and would impact their healthcare management[1]

### May be measured as:

- **Velocity** e.g. in 10MWR, 4SC, TTR, SV95C[2,3]
- **Points/score** e.g. in PUL, NSAA[4,5]

## By those affected by DMD…

**ADLs**
**Activities of daily living**

Ability to perform ADLs, maintain independence, and participate in hobbies[6]

### In qualitative evaluation of meaningful change on functional assessments:*

- Patients interpret change by tasks[6]
- A single point may reflect real functional loss[6]

*The study utilized a non-interventional, descriptive, cross-sectional qualitative design consisting of 69 semi-structured interviews with individuals with DMD and their caregivers, as well as two interviews with neuromuscular expert physiotherapists. NSAA and PUL were used to measure motor performance in ambulatory and non-ambulatory individuals, respectively.

10MWR, 10-meter walk/run; 4SC, 4 stair climb; ADL, activities of daily living; MCID, minimal clinically important difference; NSAA, North Star Ambulatory Assessment; PUL, performance upper limb; SV95C, stride velocity 95th centile; TTR, time to rise.

1. Cook CE, et al. *J Man Manip Ther*. 2008;16:E82–83; 2. Duong T, et al. *J Neuromuscul Dis*. 2021;8:939–48; 3. EMA Qualification Opinion. July 2023. Accessed February 16, 2026. https://www.ema.europa.eu/en/documents/scientificguideline/qualification-opinion-stride-velocity-95th-centile-primary-endpoint-studies-ambulatory-duchenne-muscular-dystrophy-studies_en.pdf; 4. Naarding KJ, et al. *J Neuromuscul Dis*. 2022;9:555–569; 5. Ayyar Gupta V, et al. *PLoS One*. 2023;18:e0283669; 6. Gillman A, et al. *Front Neurol. 2025;*16:1509174.

# Clinical measures can translate into meaningful outcomes for individuals with DMD




10MWR, 10-meter walk/run; 4SC, 4 stair climb; DMD, Duchenne muscular dystrophy; FVC%p, forced vital capacity percent predicted; LVEF, left ventricular ejection fraction; NSAA, North Star Ambulatory Assessment; PUL, performance upper limb; SV95C, stride velocity 95th centile; TTR, time to rise.

1. Arora H, et al. *Muscle Nerve.* 2018;58:631–638; 2. Servais L, et al. *Nat Med.* 2023;29:2391–2392; 3. McDonald CM, et al. *Lancet.* 2018;391:451–461; 4. Gillman A, et al. *Front Neurol.* 2025;16:1509174; 5. Lechner A, et al. *ERJ Open Res.* 2023;9:00176–2023; 6. Phillips MF, et al. *Am J Respir Crit Care Med.* 2001;164:2191–2194.

# What are the treatment goals for individuals with DMD and their families?



## Preserve daily function

- Maintain ability to walk, use arms and hands, and perform self-care[1,2]
- Delay loss of independence in school, work, and social life[1,2]
- Address under-recognized neurobehavioral symptoms that impact quality of life[3]

## Protect vital organs

- Prevent or delay cardiomyopathy[4]
- Preserve breathing capacity[4]

## Change the disease trajectory

- Delay irreversible milestones – loss of ambulation, need for ventilation, progressive heart and respiratory failure[1,4,5]

1. Brown V, et al. *J Patient Rep Outcomes*. 2023;7:129; 2. Gillman A, et al. *Front Neurol. 2025;*16:1509174; 3. Geuens S, et al. *J Neuromuscul Dis*. 2024;11:791–799; 4. Birnkrant DJ, et al. *Lancet Neurol.* 2018;17:251–267;
5. Duan D, et al. *Nat Rev Dis Primers*. 2021;7:13.



**Even small, measurable changes – an extra few steps, an easier rise, a longer walk, retaining upper arm strength – can make a real difference for patients and families**

# THANK YOU



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